

Contactless Cardiac Detection of Arrhythmia Based on Distribution Encoding Network With Microwave Biomedical Radar

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Abstract—As microwave sensing for cardiac monitoring advances, characterizing the population-level distributions of microwave cardiogram (MCG) and achieving robust statistical consensus are increasingly important. This article proposes a distribution-encoding technique based on dimensionality reduction that enables heatmap visualization and provides intuitive insight into the distribution of abnormal cardiac patterns. Four typical classes of arrhythmias are presented, with a particular focus on two arrhythmias: atrial premature beat (APB) and atrial fibrillation (AF). To leverage prior knowledge in distribution coordinates while automatically learning discriminative features, a distribution encoding neural network has been proposed. Validation on data from 64 patients (6115 samples) collected at two hospitals shows that both 24 GHz continuous-wave and 60 GHz frequency-modulated continuous-wave radars can effectively discriminate arrhythmias with the proposed learning mechanism. In a three-class classification (AF, APB, normal), accuracy reaches 96.15% on a small dataset of 257 samples and 90.99% on a larger dataset of 2964 samples. For common three-, four-, and five-class classifications, overall accuracies are approximately 80%, 75%, and 70%, respectively. Ablation experiments are presented to analyze the effectiveness of the learning mechanism.

Index Terms—Arrhythmia detection, biomedical radar, contactless cardiac detection, distribution encoding, neural network.

I. INTRODUCTION

DUE to advances in device miniaturization [1] and diverse clinical applications [2], contactless cardiac detection has made significant progress in the past decades [3], [4], [5], [6]. For cardiac detection, the fundamental principle is the Doppler effect: heartbeat-driven skin-surface motion modulates the phase of the electromagnetic wave, manifesting as nonlinear phase modulation [7]. Recent work on portable biomedical radars has shifted from short-term parameter estimation (e.g., heart rate and heart rate variability) [8] toward long-term healthcare applications such as disease-risk assessment [9].

Disease risk assessment requires pathological explanations and support from a substantial sample size. Detecting atrial fibrillation (AF) and other arrhythmias has become a potential research focus [10]. An approach applies transfer learning to map radar data onto electrocardiogram (ECG) [11] and enables the convenient construction of diagnostic models. However, this approach ignores the interpretability of radar signal patterns and electro-mechanical coupling between electrical activation and mechanical vibration. An alternative is to detect anomalies directly in radar signals for empirical assessment. Yet variations in subject posture, range, and inter-subject differences can alter signal patterns, complicating the extraction of stable, discriminative features for downstream classification [12].

Data-driven machine learning approaches provide new avenues for disease-risk assessment that extend beyond conventional radar signal processing. Clinical observations of anomaly-sample distributions yield two common insights. First, radar waveform manifestations of a single disease class can vary across individuals and cohorts, forming multiple subclasses. Second, different diseases present distinct waveform patterns within the same population. Capturing these distribution differences is essential for advancing the understanding of cardiac pattern generation [13].

To describe these variations, this article proposes utilizing dimensionality reduction to visualize their discrete pattern distributions as heatmaps, called distribution encoding. In this

Received 30 October 2025; revised 6 January 2026; accepted 28 January 2026. This work was supported in part by the National Key Research and Development Program of China under Grant 2022ZD0160704, in part by Shanghai Science and Technology Funds under Grant 24YL1900600, and in part by the Natural Science Foundation of China under Grant 62188102. This article is an expanded version from the 2025 IEEE MTT-S International Microwave Workshop Series on Advanced Materials and Processes for RF and THz Applications (IMWS-AMP) [DOI: 10.1109/IMWS-AMP66175.2025.11136642]. (Corresponding authors: Kang Chen; Changzhan Gu.)

This work involved human subjects or animals in its research. Approval of all ethical and experimental procedures and protocols was granted by Institutional Review Board of Ruijin Hospital, Shanghai Jiao Tong University School of Medicine under Ethics Approval Reference: (2024) Linlunshen No. 220.

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Digital Object Identifier 10.1109/TMTT.2026.3661533

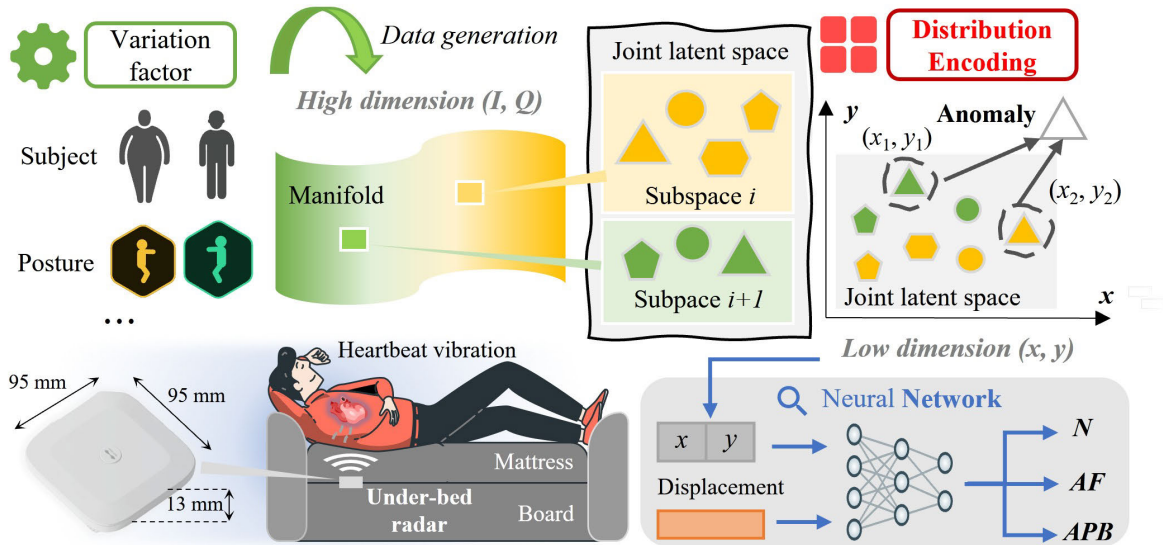


Fig. 1. Flowchart of the proposed two-stage learning mechanism.

encoding, sequence fragments from a given patient occupy a patient-specific subspace that spans multiple pattern classes. As an unsupervised method, distribution encoding offers an intuitive heatmap visualization of raw radar data while preserving prior distribution information as encoding coordinates. After understanding the signal distribution, designing a classifier based on the signal itself becomes the supervised task in the second stage. A distribution encoding neural network is then proposed to leverage prior knowledge in distribution coordinates while automatically learning discriminative features for arrhythmia classification.

Fig. 1 shows the flowchart of the two-stage learning mechanism. Differences in high-dimensional data arise due to variation factors such as subjects and postures. Low-dimensional encoding coordinates preserve the prior distribution information and are one input stream for neural network. The other input stream comprises interpretable displacement signal containing the heartbeat information, with feature extraction performed through components such as convolutional layers. Building upon previous work [10], this article mitigates network overfitting and presents results across multiclass classification and different encoding approaches, achieving a best accuracy exceeding 90% for three-class classification on datasets comprising thousands of samples. Moreover, two additional arrhythmias are incorporated, with detailed explanations provided for electro-mechanical coupling and abnormal waveforms.

This article is organized as follows. Section II sequentially introduces phase interferometry in biomedical radar, body movement interference removal, and microwave cardiogram (MCG) detection, coordinate-based distribution encoding, and neural network design. Section III sequentially demonstrates radar system design, collected datasets, and abnormal physiological signal waveforms. The heatmaps of different arrhythmias and confusion matrices under three-, four-, and five-class classifications are also detailed. Section IV experimentally validates the effectiveness of the learning

mechanism. Related data-driven learning problems are also discussed.

To our knowledge, this is an inspiring attempt to integrate unsupervised learning features into a supervised learning framework for contactless radar sensing. The contributions of this article are as follows.

- 1) The common phase interferometry in both continuous wave (CW) radar and frequency-modulated CW (FMCW) radar is illustrated, and two corresponding data acquisition platforms based on under-bed radars have been designed.
- 2) Four typical classes of arrhythmias are presented, along with an analysis of their accompanying electromechanical coupling and abnormal radar heartbeat signal waveforms.
- 3) A two-stage learning mechanism for contactless cardiac pattern classification is proposed: in the first unsupervised stage, dimensionality reduction is utilized to encode discrete pattern distributions and visualize distribution differences; in the second supervised stage, neural network automatically learns discriminative features to classify the samples.
- 4) Ablation experiments based on datasets can provide valuable References for model construction, which also inspires model development to focus on physiological interpretability.

II. PRINCIPLE AND METHOD

A. Phase Interferometry in Biomedical Radar

Short-range microwave radars are convenient to deploy and can penetrate mattresses, making under-bed monitoring less obtrusive for patients [14]. Both CW radar and FMCW radar can be deployed in under-bed scenarios. Therefore, both CW- and FMCW-based acquisition platforms have been designed to validate the effectiveness of the proposed learning mechanism.

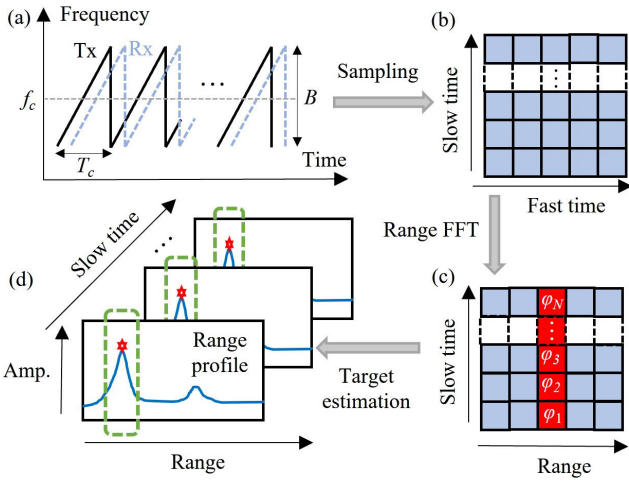


Fig. 2. Target estimation for FMCW radar. (a) Instantaneous frequency of the Tx and Rx signals. (b) Sampled FMCW chirp signal. (c) Range profile with phase variation across pulsed chirps. (d) Target estimation in each range profile based on relative amplitude.

As shown in Fig. 2, the modulated signal transmitted by the FMCW radar can be expressed as follows [15]:

$$S_{Tx}(t) = A \exp(j(2\pi f_c t + \pi \gamma t^2 + \varphi_0)), t \in \left(-\frac{T_c}{2}, \frac{T_c}{2}\right) \quad (1)$$

where t is the fast time, A is the amplitude of the signal, f_c is the center frequency, φ_0 is the initial phase of the chirp, and γ is the modulation slope, which can be denoted as $\gamma = B/T_c$, in which B is the bandwidth and T_c is the chirp repetition time (CRT). When the signal is scattered by a target, the echo signal can be expressed as follows:

$$S_{Rx}(t) = \sigma S_{Tx}(t - \Delta t) \quad (2)$$

where σ is the attenuation factor. The term Δt denotes the time delay and is given by $\Delta t = 2R(\tau)/c$, where $R(\tau) = R + v\tau$ represents the distance between the target and the radar. Here, τ is the slow time, v is the radial velocity of the target, and c is the speed of light. After de-ramping, the output intermediate frequency (IF) beat signal for a single object can be written as [14]

$$S_b(t, \tau) = \sigma A \exp\left(j\left(\frac{4\pi\gamma R(\tau)}{c}t + \frac{4\pi R(\tau)}{\lambda} + \Delta\delta\right)\right) \quad (3)$$

where λ is the wavelength, and $\Delta\delta$ is the residual phase that is negligible in a coherent system. Applying fast Fourier transform (FFT) to the fast time dimension in (3) leads to

$$S_{\text{IDFFT}}(f, \tau) = \sigma A T_c \text{sinc}(T_c(f - f_b)) \exp\left(j\frac{4\pi R(\tau)}{\lambda}\right) \quad (4)$$

where $f_b = 2\gamma R(\tau)/c$ is the beat frequency. As shown in Fig. 3(a), the phase history, which represents the displacement of the moving target, is reconstructed by extracting the equivalent I/Q signal from the corresponding target range bin m_0 . Given that f_0 (close to f_b) is the frequency corresponding to the chosen range bin m_0 , substituting f_0 into (4) leads to

$$S_{\text{IDFFT}}(f_0, \tau) = \sigma A T_c \text{sinc}(T_c(f_0 - f_b)) \exp\left(j\frac{4\pi R(\tau)}{\lambda}\right). \quad (5)$$

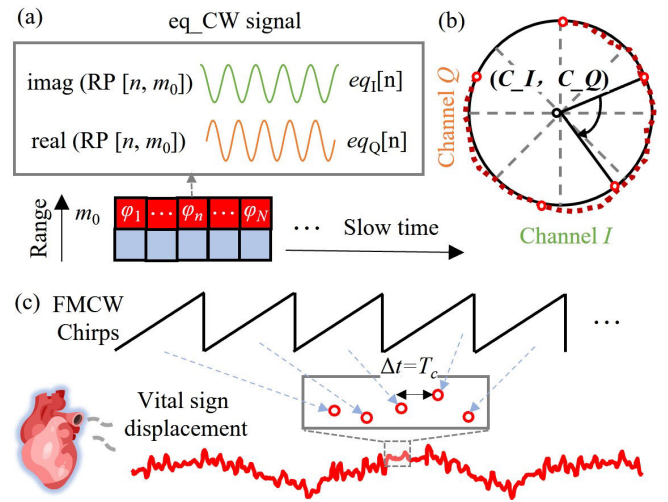


Fig. 3. Each chirp converts to a single sampled point in the extracted equivalent I/Q signals. (a) Equivalent CW signal is extracted from each range profile at the slow time dimension. (b) Equivalent CW signals undergo circle fitting similar to CW signals. (c) Equivalent sampling time is the CRT.

From (5), it can be seen that the target interferometric phase change at the target range bin originates from the slow time dimension τ , and proximity stationary clutters equivalently result in a direct current (DC) bias at the target range bin [5]. Since the human body and under-bed radar are positioned in a relatively fixed configuration, the position of the human body can be determined solely on the maximum amplitude of the range profile between the mattress thickness and the maximum monitoring range

$$m_0 = \arg \max_m \text{abs}(\text{RP}[n, m]), r_{\text{mattress}} \leq R(m) \leq r_{\text{maximum}} \quad (6)$$

where m_0 is the target range bin index, n is the chirp repetition index, $\text{abs}(\text{RP}[n, m])$ is the amplitude of range profile, r_{mattress} is the thickness of mattresses like 20 cm, and r_{maximum} is the maximum effective monitoring range like 100 cm. The sampled equivalent I/Q signals can be expressed as follows:

$$eq_I[n] = A_{\text{eq}}(n) \cdot \cos\left(\frac{4\pi \cdot R_{i0}[n]}{\lambda}\right) + C_I(n) \quad (7)$$

$$eq_Q[n] = A_{\text{eq}}(n) \cdot \sin\left(\frac{4\pi \cdot R_{i0}[n]}{\lambda}\right) + C_Q(n) \quad (8)$$

where $A_{\text{eq}}(n)$ is the equivalent amplitude, $R_{i0}[n] = d_{i0} + x_{i0}[n]$ is the absolute distance of the target with d_{i0} being the mean distance of the moving target and $x_{i0}[n]$ being the target displacement motion to be detected, and $C_I(n)/C_Q(n)$ are equivalent DC offsets. In FMCW radar, n corresponds to the chirp repetition index at the slow time dimension τ , and the equivalent sampling time is the CRT as shown in Fig. 3(c).

A typical form of the emitted signal of CW is as follows [16]:

$$T(t) = A_T \cos(2\pi f t + \varphi(t)) \quad (9)$$

where f is the carrier frequency, A_T is the amplitude, and $\varphi(t)$ is the phase noise generated by the signal generator. The received echo is as follows:

$$R(t) = A_R \cos\left[2\pi f t - \frac{4\pi d_o}{\lambda} - \frac{4\pi x(t)}{\lambda} + \varphi\left(t - \frac{2d_o}{c}\right)\right] \quad (10)$$

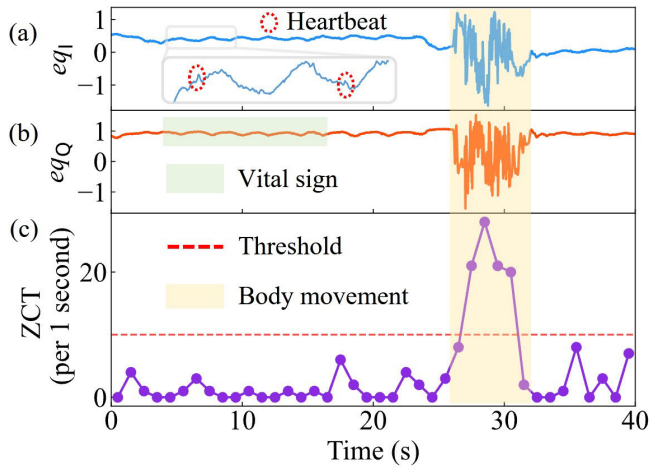


Fig. 4. Equivalent I/Q signals and ZCT values from the FMCW radar-based post-synchronous acquisition platform. (a) Equivalent I signal. (b) Equivalent Q signal. (c) ZCT values tend to be high when body movement occurs.

where A_R is the amplitude of the echo, d_0 is the nominal distance, $x(t)$ is the vibration, c is the speed of light, and λ is the wavelength. For a quadrature down-conversion radar transceiver structure, the radio frequency (RF) signal is directly converted to the baseband I/Q outputs. The sampled I/Q outputs are as follows:

$$B_I[n] = A(n) \cdot \cos \left[\frac{4\pi x(n)}{\lambda} + \Delta\varphi \right] + DC_I(n) \quad (11)$$

$$B_Q[n] = A(n) \cdot \sin \left[\frac{4\pi x(n)}{\lambda} + \Delta\varphi \right] + DC_Q(n) \quad (12)$$

where $A(n)$ is the amplitude of the measured CW signal, $\Delta\varphi$ is the residual phase, and $DC_I(n)/DC_Q(n)$ are the DC offsets of the I/Q channels. In CW radar, n corresponds to the sampling index within a continuous waveform, and the sampling time depends on the analog-to-digital converter (ADC) settings.

B. Body Movement Interference Removal

As shown in Fig. 4, body movement interference signals lack periodicity and exhibit intense instantaneous fluctuations when compared to vital sign signals. Zero-crossing times (ZCTs) are a sensitive indicator reflecting body movement and has been widely adopted in speech recognition, which can quantify the zero-crossing characteristic of the signal [17]. By calculating the ZCT value and setting the threshold, the recognition of the body movement can be realized. ZCT is defined as follows:

ZCT

$$= \sum_{k=1}^{N-1} \prod (X(k)X(k+1) < 0 \text{ and } |X(k) - X(k+1)| \geq \epsilon) \quad (13)$$

where $\prod(\cdot)$ means the function that returns one when the condition is true and zero when it is false, ϵ is a threshold to avoid miscounting zero crossing due to noise, $X(k)$ is the digital signal to be analyzed, and N is the sample number of digital signal. In this work, ZCT is employed to filter out body

movement segments, ensuring the reliability of subsequent model training and validation.

As shown in Fig. 4(c), ZCT values are calculated from the equivalent I/Q signals [17], and ZCT values tend to be high when body movement occurs. Reliable vital signs unaffected by body movement are selected for high-confidence model validation, and I/Q signals with high ZCT values are not included for the experiments. Future work can address signal confidence assessment under movement interference and integrate techniques for movement cancellation, including multiple radars from different angles [4], empirical modal decomposition (EMD) [18], and so on [19].

C. MCG Detection

The heartbeat is a periodic movement that is faster and more vigorous than respiration. An intuitive insight is that kinetic waveforms, such as velocity or acceleration, can provide a more vivid representation of cardiac activity [20]. By enhancing the heartbeat waveform from displacement signal, MCG can be obtained.

As shown in Fig. 3(b), circle fitting is commonly employed to remove the unwanted DC components ($DC_I(n)/DC_Q(n)$ in CW radar and $C_I(n)/C_Q(n)$ in FMCW radar) and recover the displacement motion from the phase. The relationship between the recovered displacement motion and phase can be expressed as follows:

$$\theta(t) = \frac{4\pi}{\lambda} x(t). \quad (14)$$

A linear coefficient can be found between phase $\theta(t)$ and displacement $x(t)$. For a resting subject, the vital sign can be regarded as the superposition of the periodic respiratory signal $x_r(t)$ and the heartbeat signal $x_h(t)$

$$x(t) = x_r(t) + x_h(t). \quad (15)$$

Phase demodulation techniques are then employed to recover the displacement [16], [21], and accurate phase accumulation (APA) is one of these techniques

$$x[n] = \frac{\lambda}{4\pi} \sum_{k=2}^n \text{atan} 2 \frac{I[k-1]Q[k] - I[k]Q[k-1]}{I[k-1]I[k] + Q[k]Q[k-1]}. \quad (16)$$

For vital sign, the slowly varying respiratory component constitutes the trend component of the displacement signal. After differentiation, the trend component can be attenuated, while the rapidly pulsating heartbeat component can be amplified. The differential displacement is as follows:

$$d[n] = x[n] - x[n-1], \quad n = 1, 2, 3 \dots \quad (17)$$

where $x[0]$ is set as zero to keep the signal length unchanged. The differential displacement signal physically represents the concept of velocity. Although differential displacement keeps more dynamic information, it is more susceptible to noise. Applying Savitzky-Golay filter can help smooth the waveform for improved heartbeat visualization. The window time is defined as follows:

$$T = L/f_s \quad (18)$$

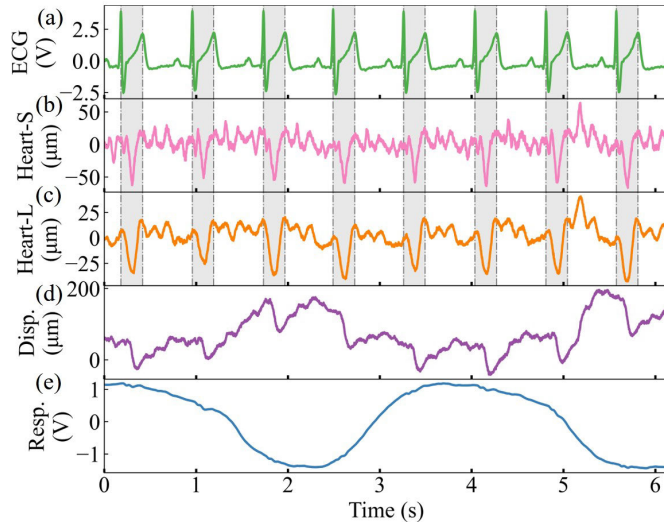


Fig. 5. MCG detection for the vital sign from the CW radar-based synchronous acquisition platform. (a) ECG signal. (b) Heart extraction by short window time of 0.06 s. (c) Heart extraction by long window time of 0.18 s. (d) Displacement signal. (e) Respiration (resp.) belt signal.

where T is the window time, L is the length of the filter window, and f_s is the signal sampling rate. The smoothing formula for its data points is as follows:

$$d^*[n] = \sum_{j=-m}^m c_j d[n+j] \quad (19)$$

where $2m+1$ means the window length, and c_j is obtained by performing a k -order polynomial fit on the raw data. The Savitzky-Golay filter has advantages in preserving the original waveform while reducing noise, and it is employed for heartbeat extraction following displacement differential operations.

Fig. 5 shows the vital sign acquisition results from a healthy individual at a sampling rate of 500 Hz. During signal processing, set the polyorder parameter of the Savitzky-Golay filter to two. Two windowing times, 0.06 s and 0.18 s, are set for waveform smoothing. The corresponding filter window lengths can be calculated based on the sampling rate. The longer the window, the smoother the signal becomes, but some detail is also lost. Comparing Fig. 5(a) to (c), it can be observed that the peak intervals of the MCG exhibit periodic consistency with those of the ECG. Comparing Fig. 5(d) and (e), it is observed that the displacement (disp.) signal is slowly and periodically modulated by respiration.

D. Electro-Mechanical Coupling

As shown in Fig. 6, during synchronous acquisition, the relationship of electro-mechanical coupling is depicted. Atrial contraction is generally considered the initiation of a cardiac cycle, during which the nerve impulse is generated in the atria. At this point, the ECG shows the generation of the P wave. The activity of atrial contraction is relatively weak compared to that of ventricular contraction, so the subsequent mechanical activity peak is not pronounced.

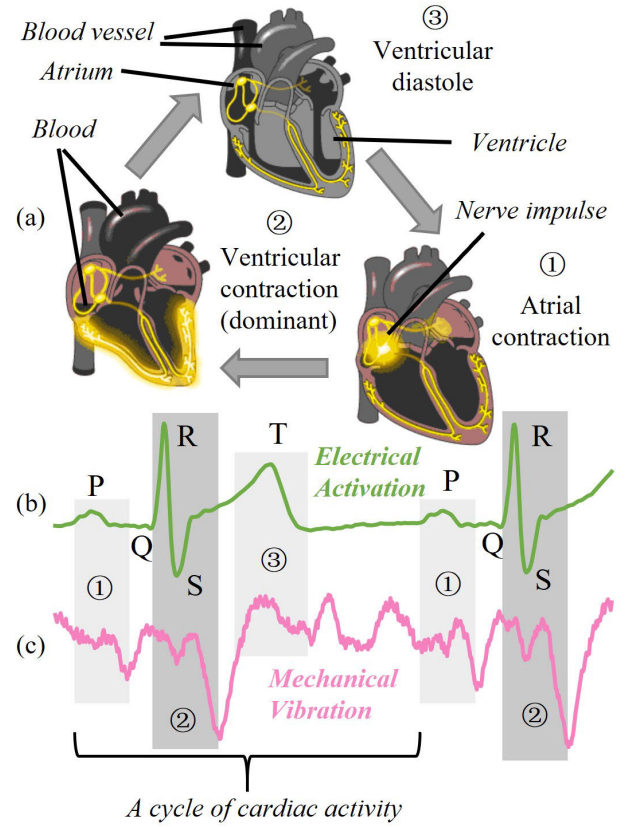


Fig. 6. Electromechanical coupling. (a) Periodic activity of the heartbeat. (b) Periodic electrical activation. (c) Periodic mechanical vibration.

The dominant event in cardiac activity is ventricular contraction. At this point, the generation of QRS complexes can be observed on the ECG. The relative relationship between the midpoints of the QRS complexes reflects the ventricular contraction capacity. However, due to variations in the structure of cardiac chambers among individuals, the time delay associated with the transmission of electrical activation-induced mechanical vibration can differ. However, within a short timeframe, this delay remains constant. Consequently, consistent peak-to-peak cycles can be observed between the MCG and ECG. Various methods for measuring the heartbeat intervals in MCG focus on capturing the main peak in this mechanical vibration [22].

Finally, ventricular diastole occurs. In MCG recordings, this phase typically presents as a relatively flat interval compared to systole, during which a negative peak opposite the main peak can be observed. Correspondingly, T waves indicating ventricular repolarization can be observed on the ECG. Electro-mechanical coupling reflects the physiological interpretability of MCG, which forms the theoretical basis for its application in monitoring disordered cardiac activities such as arrhythmia.

E. Coordinate-Based Encoding of Distribution

Different individuals possess unique distributions of respiratory and heartbeat patterns, which collectively form a distinct subspace for each individual. When these subspaces

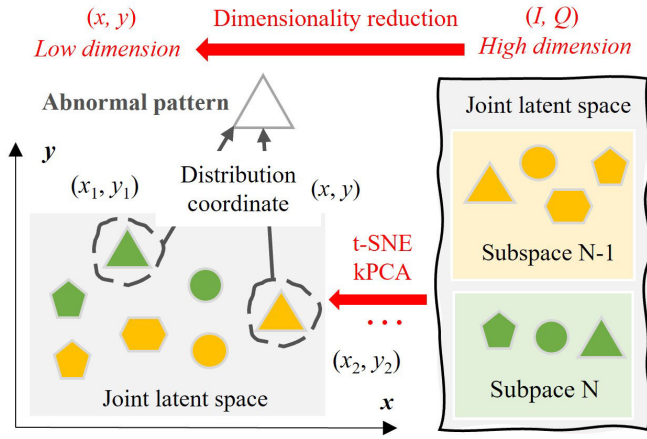


Fig. 7. Coordinate-based encoding of arrhythmia segment distribution when the encoding dimension is 2.

are combined, they form a joint latent space as illustrated in Fig. 7. Dimensionality reduction is utilized to extract key feature dimensions from this high-dimensional latent space while maintaining the proximity between the original data segments. Kernel principal component analysis (kPCA) [23] and t-distributed stochastic neighbor embedding (t-SNE) [24] are two of the most popular nonlinear dimensionality reduction approaches in machine learning.

The core idea of kPCA lies in the kernel trick, which realizes an implicit mapping of the original data into a nonlinear feature space, where linear PCA is then performed. Because kernels can introduce nonlinearity, the resulting process corresponds to nonlinear dimensionality reduction in the original data space. The kernel function can be expressed as follows:

$$k(x_i, x_j) = \langle \Phi(x_i), \Phi(x_j) \rangle \quad (20)$$

where x_i and x_j are two data points in the original data space, Φ is the mapping function that maps the original data points to a high-dimensional space, and $k(\cdot, \cdot)$ is the kernel function. The core parameter of kPCA is the selected kernel type.

Probabilistic consistency is emphasized in t-SNE to preserve the local distribution structure by aligning high-dimensional points with their low-dimensional counterparts. The low-dimensional map of t-SNE is a t-distribution with a single degree of freedom, expressed as follows:

$$q_{ij} = \frac{(1 + \|y_i - y_j\|^2)^{-1}}{\sum_{k \neq l} (1 + \|y_k - y_l\|^2)^{-1}}, y_i \in \mathbf{R}^N \quad (21)$$

where q_{ij} is the similarity between two data points y_i and y_j in the low-dimensional space, and N is the dimensionality set as 2. The probability distribution Q in a low-dimensional space is constituted by q_{ij} . The Kullback–Leibler (KL) divergence between the high-dimensional distribution P and the low-dimensional distribution Q is minimized by optimizing the low-dimensional coordinates Y . The minimization of KL divergence is as follows:

$$\min C(Y) = \min \text{KL}(P||Q) = \min \sum_{i \neq j} p_{ij} \log \frac{p_{ij}}{q_{ij}} \quad (22)$$

where $Y = \{\dots, y_i, y_j, \dots\}$ means all the data points in the low-dimensional space. The probability distribution P in the high-dimensional space can be calculated as follows:

$$p_{ji} = \frac{\exp(-\|x_i - x_j\|^2 / 2\sigma_i^2)}{\sum_{k \neq i} \exp(-\|x_i - x_k\|^2 / 2\sigma_i^2)} \quad (23)$$

$$p_{ij} = \frac{p_{ji} + p_{ilj}}{2M} \quad (24)$$

where x_i and x_j are two data points in the original data space, σ_i^2 is the variance of the Gaussian distribution centered at the data point x_i , p_{ij} is the probability of neighbor embedding, and M is the total number of data points.

By employing dimensionality reduction approaches, the reduced 2-D space corresponds to an (x, y) Cartesian coordinate system, facilitating intuitive visualization. Encoding the distribution serves as a discrete representation, enhancing the inference capability of downstream machine learning models for tasks such as classification [25].

However, due to the global optimization of t-SNE, this leads to certain deployment challenges. A general approach is to employ parametric t-SNE, which can be achieved by training a corresponding multivariate regressor or lightweight multilayer perceptron [26]. For the weights W of the multilayer perceptron, the gradient descent process proceeds as follows:

$$\frac{\delta C(Y)}{\delta W} = \frac{\delta C(Y)}{\delta f(x_i|W)} \frac{\delta f(x_i|W)}{\delta W} \quad (25)$$

where $\delta f(x_i|W)/\delta W$ is computed by backpropagation, and $\delta C(Y)/\delta f(x_i|W)$ is defined by a formula of $f(x_i|W)$. In parametric t-SNE, the parametric mapping from the data space X to the low-dimensional space Y is parametrized by means of a feed-forward neural network with weights W [26]. Therefore, a mapping function can be obtained by parametric t-SNE while preserving the effectiveness of t-SNE.

F. Distribution Encoding Network

Based on the proposed coordinate-based distribution encoding, a distribution encoding network has been specifically designed. The neural network backbone adheres to the principles of classical encoder–decoder architecture, which has proven effective in sequence modeling [27]. The neural network employs a combination of residual connection [28] and transformer encoder [29] to extract features from vital sign displacement, aiming to enhance the extraction of nonlinear features in this branch. The convolution kernel applied to the convolutional layer is 1×3 , with the ReLU activation function [30]. The transformer encoder consists of four attention heads. The dropout [31] is employed in two convolutional layers to mitigate overfitting, and the dropout probability is set as 0.4.

The neural network structure is shown in Fig. 8. A 10-s time window is utilized for signal segmentation, where the I/Q signals are mapped into (x, y) coordinates, and the demodulated displacement signal is utilized as another branch of the network input. The 2-D coordinate encoding is aligned to a 32-D tensor by linear transformations. The original 500-D vital sign displacement is also aligned to a 32-D tensor after feature extraction. These two 32-D tensors are contacted to

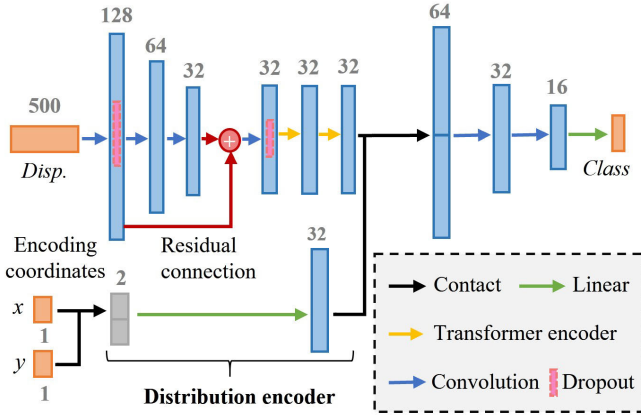
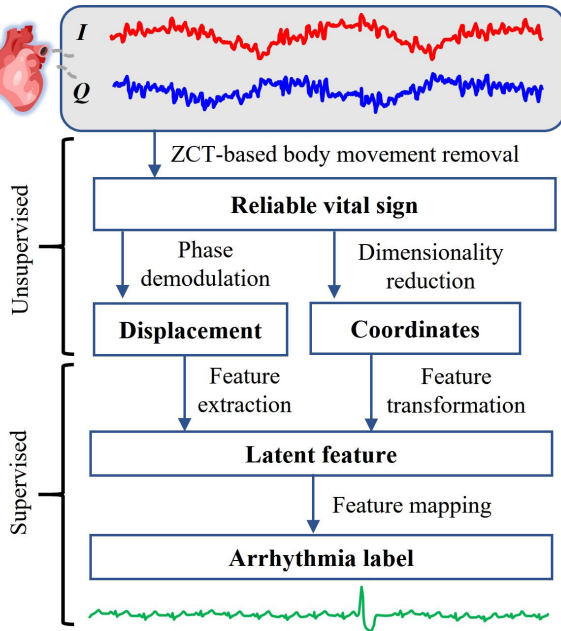


Fig. 8. Proposed distribution-encoding network.

Fig. 9. Two-stage learning mechanism from I/Q signals to label.

a 64-D tensor. The above process constitutes the distribution encoder. Finally, the 64-D tensor is decoded to match the dimension of the class label. The loss function adopted is the typical cross-entropy (CE) function as follows:

$$\text{CE Loss} = - \sum_{i=1}^C r_i \cdot \log(\hat{r}_i) \quad (26)$$

where C is the total number of classes, r_i is the true label tensor, and $\hat{r}_i$ is the predicted label tensor.

The process flowchart from I/Q signals to label is shown in Fig. 9. It consists of two stages: unsupervised and supervised. In the unsupervised stage, interpretable displacement is demodulated, which incorporates heartbeat information. At the same time, the raw radar signals, after dimensionality reduction, preserve the similarity of their original distribution, a feature referred to as distribution coordinates. In the supervised stage, neural networks are leveraged for their powerful nonlinear capabilities. Through feature extraction and fusion,

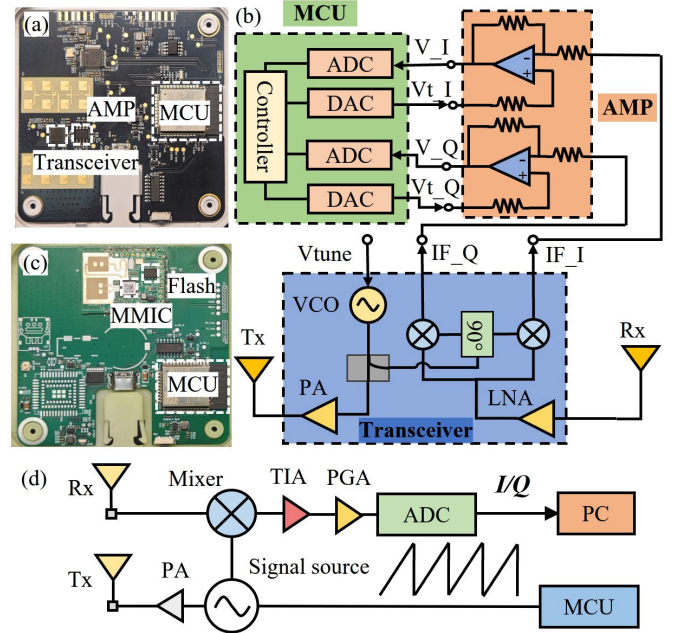


Fig. 10. Platform for under-bed radar monitoring. (a) Custom-designed 24 GHz CW radar. (b) Structure of CW radar. (c) Custom-designed 60 GHz FMCW radar. (d) Structure of FMCW radar.

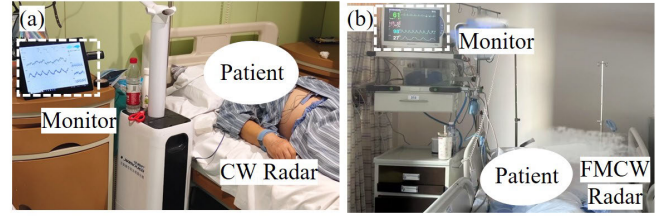


Fig. 11. Monitoring scenario. (a) Synchronous data collection at Shanghai Sixth People's Hospital. (b) Postsynchronous data collection at Ruijin Hospital.

the distribution encoding network completes classification tasks aligned with the labels. This two-stage process endows the model with overall physiological interpretability.

III. EXPERIMENT AND RESULTS

A. Radar System Design

In Fig. 10, two consumer-designed radars and their structures are shown. Fig. 10(a) depicts the outlook of a 24 GHz CW radar, and Fig. 10(c) depicts the outlook of a 60 GHz FMCW radar. Fig. 11(a) shows the data-collection setup at Shanghai Sixth People's Hospital, and Fig. 11(b) shows the data-collection scenario at Ruijin Hospital. All participants involved in the process consented to the experiment. The printed circuit board (PCB) design allows all electronic components to be integrated on a 95×95 mm surface, and the enclosure has a thickness of only 13 mm. In practice, the radar sensors are positioned attached to the bed board and beneath the bed mattress.

Both FMCW and CW radars can be adopted for vital sign monitoring, and static clutter can be treated as equivalent to the DC component obtained in the I/Q circle fitting. FMCW radar

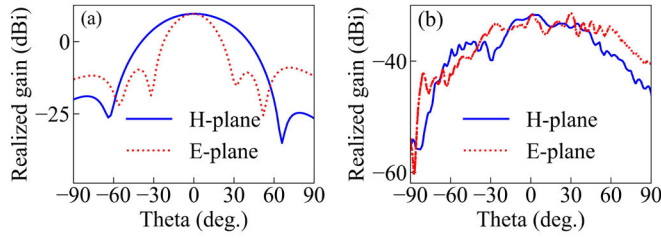


Fig. 12. Radiation pattern of the antenna. (a) CW multiple-patch antenna. (b) FMCW single-patch antenna.

TABLE I
MAIN PARAMETERS IN THE CW RADAR

Parameter	Value
Tx power	9 dBm
Baseband amplifier gain	3 dB
Sampling rate	50 Hz
Carrier frequency	24 GHz
Antenna gain	13 dBi
Antenna beamwidth	E plane: $\pm 12^\circ$ H plane: $\pm 23^\circ$

can measure distance and better distinguish monitored objects at different distances, making it suitable for multiperson scenarios. The specific product models of FMCW radar are the Hecaray SA-05 and SA-02-R. In contrast, CW radar imposes stricter requirements on single-person monitoring scenarios, despite its advantages of simple structure and rapid sampling rate [19]. As shown in Fig. 10(b), due to the direct conversion architecture, the baseband output signals contain DC offsets that may saturate the amplifiers (AMPs). Therefore, a DC tuning algorithm is written in the microcontroller unit (MCU) to adaptively adjust the DC offsets to make full use of the dynamic range of the AMPs [6]. DC-tuning utilizes MCU to control the digital-to-analog converter (DAC), forming a feedback loop. When environmental changes cause DC shifts, the MCU outputs V_{t_I} and V_{t_Q} as feedback loop signals, thereby ensuring that the output $V_{_I}$ and $V_{_Q}$ are fully amplified. DC tuning enables the system to operate more robustly in cluttered scenes.

For the custom-designed CW radar system, a 24 GHz transceiver SRK1102A is utilized, and the baseband AMP is LTC8552. The main parameters in the CW radar are shown in Table I. The Tx power is set as 9 dBm, and the gain of baseband AMP is 3 dB. The antenna beamwidth is $\pm 12^\circ$ in *E*-plane and $\pm 23^\circ$ in *H*-plane due to the multiple-patch antenna, and the antenna radiation pattern is shown in Fig. 12(a).

For the custom-designed FMCW radar system, the ADT6101P, a 60 GHz radar system-on-chip (SoC), is utilized. This SoC integrates the complete RF transceiver signal chain and ADCs, while custom-designed PCB antennas are employed for signal transmission and reception. The main parameters in the FMCW radar are shown in Table II. The Tx power is set as 12 dBm, and the gains of transimpedance AMP (TIA) and programmable gain AMP (PGA) are 0 and 40 dB, respectively. The sweeping bandwidth is from 57 to 59.5 GHz

TABLE II
MAIN PARAMETERS IN THE FMCW RADAR

Parameter	Value
Tx power	12 dBm
TIA gain	0 dB
PGA gain	40 dB
PGA bandwidth	100 kHz ~ 2 MHz
Sampling rate	7.14 MHz
Sampling points in a chirp	128
Chirp repetition time	42 μ s
Chirps in a frame	32
Sweeping bandwidth	57 GHz ~ 59.5 GHz
Antenna gain	6 dBi
Antenna beamwidth	E plane: $\pm 50^\circ$ H plane: $\pm 50^\circ$

TABLE III
PATIENTS OF THE COLLECTED DATASETS

Class	N	APB	VPB	AVB	AF
Six-Imbalance	3	3	14	10	22
Six-Subset	3	3	3	3	3
Ruijin			12		
Total			64		

TABLE IV
SAMPLES OF THE COLLECTED DATASETS

Class	N	APB	VPB	AVB	AF
Six-Imbalance	84	79	666	380	1,100
Six-Subset	84	79	75	72	94
Ruijin	1,239	747	757	85	978
Total			6,115		

for a bandwidth of 2500 MHz. The chip features two transmit and two receive (2T2R) channels for a total of four channels, but in practice, only one transmit and one receive channel (1T1R) is utilized. The antenna beamwidth is $\pm 50^\circ$ due to the single-patch antenna, and the antenna radiation pattern is shown in Fig. 12(b).

B. Dataset

Four arrhythmias are selected as typical examples: atrial premature beat (APB), ventricular premature beat (VPB), AF, and atrioventricular conduction block (AVB), with atrial and VPBs representing two subclasses within the premature beat (PB) category. During the process, normal (N) sinus heartbeats are also collected as a control. The dataset collection platforms are illustrated in Fig. 11. The specific numbers of patients and samples are shown in Tables III and IV, respectively.

As shown in Fig. 11(a), for CW radar, data acquisition is conducted in the presence of a nurse to ensure that the single-subject monitoring condition is minimally susceptible to surrounding interference. The data are collected synchronously using instrumentation that leverages the MCU on the radar board to sample both the ECG sensor channel and the two *I/Q* radar channels. Data acquisition is presented as a locally

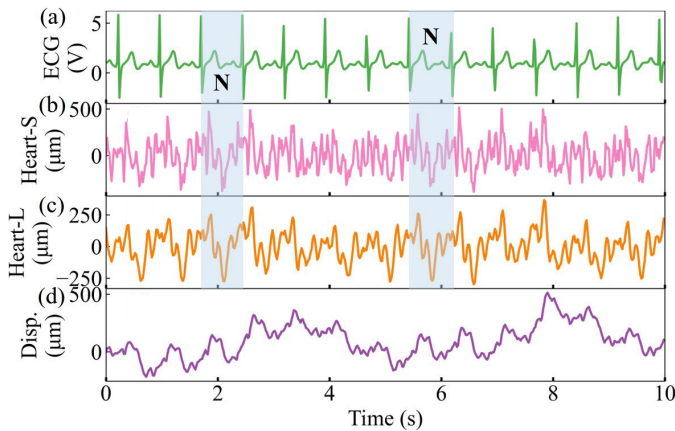


Fig. 13. Vital sign of normal (N) sinus heartbeats from the CW radar-based synchronous acquisition platform. (a) ECG signal. (b) Heart extraction by short window time of 0.06 s. (c) Heart extraction by long window time of 0.18 s. (d) Displacement signal.

synchronous acquisition platform. Utilizing the 24 GHz CW radar on the synchronous acquisition platform, data are collected from 52 relevant patients, yielding 2309 samples. Sample labels are derived from locally synchronous ECG signals. Although the number of patients is relatively large, the number of effective samples is limited because each patient is screened for only 5 min.

As shown in Fig. 11(b), for FMCW radar, the monitor and radar form a postsynchronized acquisition platform. The FMCW radar automatically locates the most likely distance bin for the target, thereby reducing clutter interference. This nurse-unattended acquisition enhances data collection efficiency but also presents certain postsynchronization challenges, including device matching and the digitization of monitor data. Utilizing the 60 GHz FMCW radar on the postsynchronous acquisition platform, data are collected from 12 relevant patients, resulting in 3806 samples. Patients undergo extended day–night monitoring, with labels obtained from postsynchronous ECG signals and manual retrospective annotation of diurnal data. Each patient is monitored for multiple consecutive days, so despite the smaller number of patients, the effective sample count is substantial. All I/Q signals are sampled at 50 Hz, and each dataset sample lasts 10 s, giving a fixed sample length of 500 sample points.

C. Physiological Signal Waveforms

The physiological signal waveforms presented in this article are from the CW radar-based synchronous acquisition system, serving as standard examples. The heartbeat extraction method follows the procedure depicted in Fig. 5, utilizing short-term and long-term time windows of 0.06 and 0.18 s, respectively. The signals are sampled at 50 Hz, resulting in Savitzky–Golay filter window lengths of 3 and 9, respectively. Fig. 13 illustrates the vital sign of normal sinus heartbeats, characterized by regular periodicity and a high degree of consistency in waveform details between heartbeats. Fig. 14 illustrates the vital sign of AF, where the heartbeat cycle is irregular, leading to corresponding irregular waveform disturbances in both

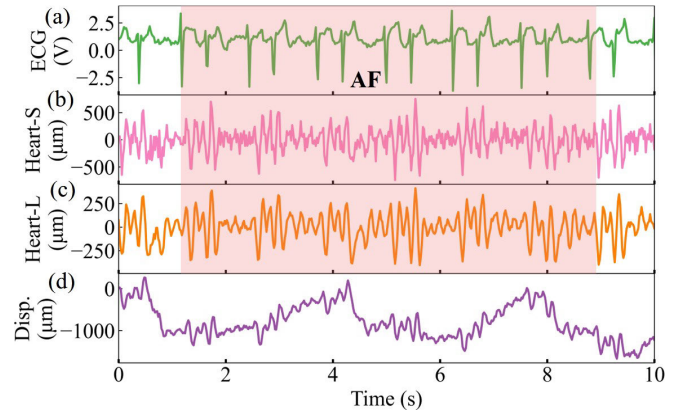


Fig. 14. Vital sign of AF from the CW radar-based synchronous acquisition platform. (a) ECG signal. (b) Heart extraction by short window time of 0.06 s. (c) Heart extraction by long window time of 0.18 s. (d) Displacement signal.

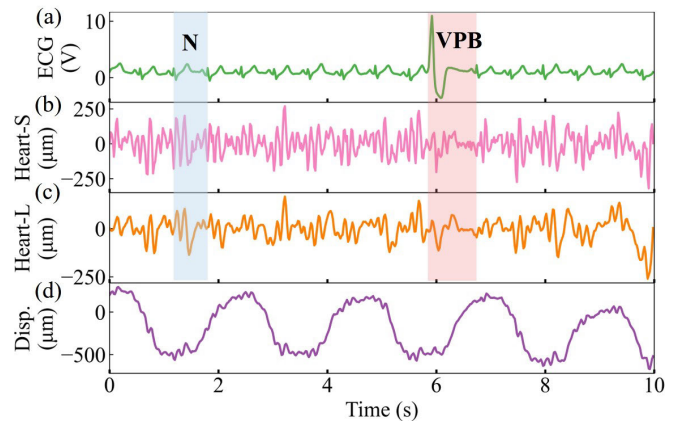


Fig. 15. Vital sign of VPB from the CW radar-based synchronous acquisition platform. (a) ECG signal. (b) Heart extraction by short window time of 0.06 s. (c) Heart extraction by long window time of 0.18 s. (d) Displacement signal.

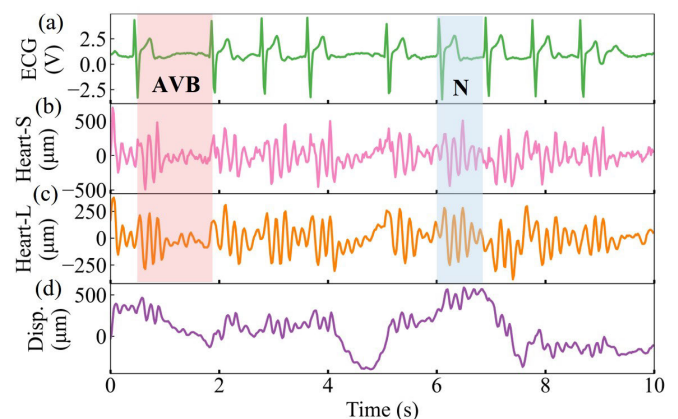


Fig. 16. Vital sign of AVB from the CW radar-based synchronous acquisition platform. (a) ECG signal. (b) Heart extraction by short window time of 0.06 s. (c) Heart extraction by long window time of 0.18 s. (d) Displacement signal.

MCG and ECG signals. Fig. 15 illustrates the vital sign of VPB, in which a PB disrupts the regularity of the heartbeat, and waveform details differ from those of normal heartbeats in both MCG and ECG. Fig. 16 illustrates the vital sign of

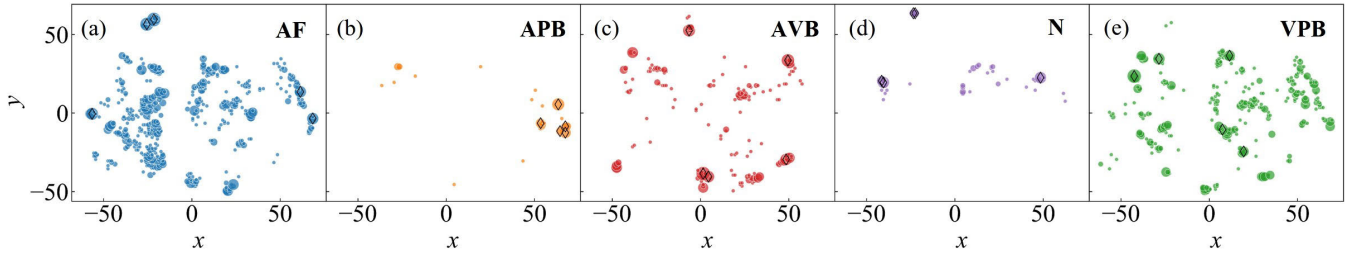


Fig. 17. Distributions of dataset Six-Imbalance with 52 patients and 2309 samples by t-SNE. (a) Subjects with AF. (b) Subjects with APB. (c) Subjects with AVB. (d) Normal subjects. (e) Subjects with VPB. The five small diamonds in the diagram represent the five subcenters with the highest frequency.

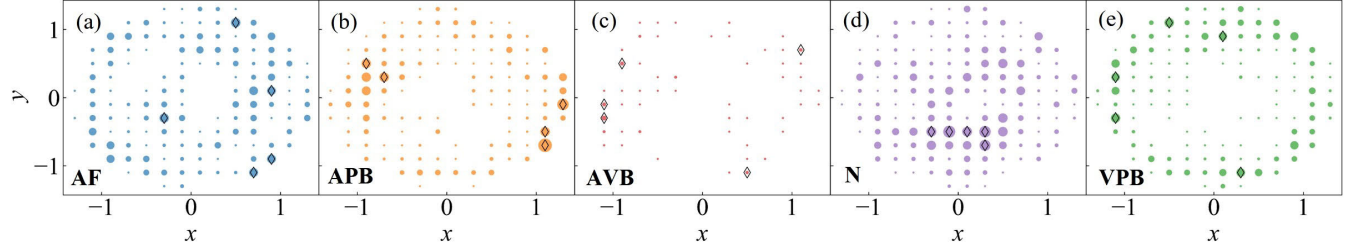


Fig. 18. Distributions of dataset Ruijin with 12 patients and 3806 samples by kPCA. (a) Subjects with AF. (b) Subjects with APB. (c) Subjects with AVB. (d) Normal subjects. (e) Subjects with VPB. The five small diamonds in the diagram represent the five subcenters with the highest frequency.

atrioventricular conduction block, where prolongation of the P–R interval interrupts the periodicity of the heartbeat, and waveform details also differ from normal heartbeats in both MCG and ECG.

The previous method has distinguished AF in the frequency domain by analyzing RR intervals [6], but the vital sign waveforms contain more detailed pathological information as shown in Fig. 14. Therefore, this article attempts to classify arrhythmias by analyzing the signals in the morphological domain and aims to build an arrhythmia classification model based on vital sign displacement.

D. Encoding Coordinate Visualization

Data are collected from 64 patients, categorized into four typical classes of arrhythmias. By segmenting the collected radar signal with a window length of 10 s, 6115 segments are obtained, equivalent to 1019 min of data. These 10-s I/Q signal segments are encoded into a 2-D distribution coordinate (x, y) utilizing dimensionality reduction. The coordinates obtained from the dimensionality reduction are continuous, and a simple frequency count is performed for visualization. For the grid at (i, j) , the corresponding frequency h_{ij} is calculated as follows:

$$h_{ij} = \frac{\text{count} \left\{ (x, y) | x \in \left(i - \frac{H}{2}, i + \frac{H}{2} \right), y \in \left(j - \frac{H}{2}, j + \frac{H}{2} \right) \right\}}{\sum \text{count} \{ (x, y) \}} \quad (27)$$

where i and j are discrete integer values, and H is the grid size. Since the distribution coordinates are obtained as a low-dimensional embedding that considers the joint latent space, the distribution encoding preserves the global prior information accordingly.

Distributions of dataset Six-Imbalance with 52 patients and 2309 samples by t-SNE are shown in Fig. 17. The perplexity is set to 10, and the maximum number of iterations is set to

1000. The grid size is 1. Distributions of dataset Ruijin with 12 patients and 3806 samples by kPCA are shown in Fig. 18. The kernel function employed is a polynomial kernel. The grid size is 0.2. The relative size of the grid points indicates the relative frequency values, and the five small diamonds in the diagram represent the five subcenters with the highest frequency. Across different arrhythmia classes, the subcenters obtained via t-SNE or kPCA are distinct and separated from normal samples.

E. Neural Network Classification Results

A distribution-encoding network has been designed to utilize the global prior information in distribution coordinates and to perform nonlinear feature extraction of vital sign displacement. A subdataset Six-Subset consisting of 15 subjects, with three subjects each from N, AF, APB, VPB, and AVB, includes a total of 404 segments. The samples for the sub-dataset are shown in Table IV. The validation of this sub-dataset corresponds to the small sample sizes commonly encountered in practical scenarios while avoiding the impact of sample imbalance on classifier learning. The sub-dataset is randomly divided into training, validation, and testing sets with a ratio of 6:2:2. The training batch size is set to 32, and the model is trained on an RTX 3090 graphics-processing unit (GPU). The learning rate is $5e-5$. During the data training process, the early stopping [32] approach is employed to prevent overfitting to the training set, which means stopping training when the validation loss stops decreasing. The patience of early stopping is set to 10.

For the testing set, confusion matrices in percentage format on the balanced dataset Six-Subset are shown in Fig. 19. Classification accuracy for the three-, four-, and five-class problems is 96.15%, 77.61%, and 68.29%, respectively. It can be observed that as the number of classes increases,

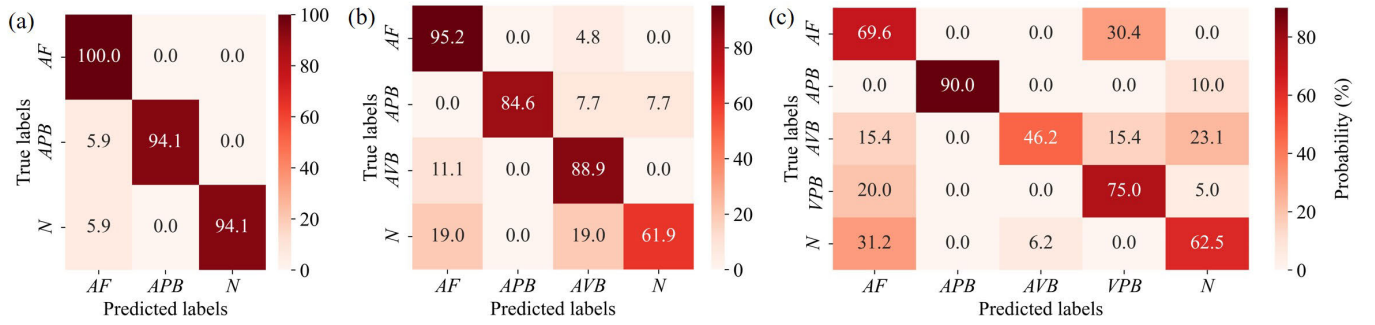


Fig. 19. Confusion matrices in percentage format on balanced dataset Six-Subset. (a) Three-class classification of AF-APB-N. (b) Four-class classification of AF-APB-AVB-N. (c) Five-class classification of AF-APB-AVB-VPB-N.

TABLE V

MULTI-CLASS CLASSIFICATION RESULTS ON DATASET SIX-SUBSET

Number of classes	Acc.	Pre.	Rec.	F1	Kappa
3	0.9615	0.9667	0.9608	0.9623	0.9422
4	0.7761	0.8036	0.7711	0.7728	0.6951
5	0.6829	0.7483	0.6864	0.7029	0.5902

the accuracy gradually decreases. This indicates that learning multiclass abnormal arrhythmia patterns presents certain challenges. Five metrics—accuracy (Acc.), precision (Pre.), recall (Rec.), F1-score (F1), and Kappa—are selected to quantify the learning effectiveness of the classifier [33]. Accuracy is suitable for evaluating overall correctness in balanced datasets. Precision and recall, in contrast, reflect the effectiveness of the model in reducing false positives and minimizing false negatives, respectively. F1-score and Kappa are more appropriate for assessing overall performance on imbalanced datasets. The F1-score combines precision and recall, while Kappa measures the performance beyond chance-level guessing. The chosen encoding strategy is t-SNE, with an encoding dimension of 2. The multiclass classification results on dataset Six-Subset are shown in Table V.

IV. ABLATION AND DISCUSSION

For machine learning, it is beneficial to provide detailed explanations of the learning approaches applied to datasets; this aspect is referred to as ablation experiments. Because the distribution-encoding network introduces prior distribution information through the encoding coordinates, the encoding strategy and dimension are discussed for their impact on the learning results. Parametric t-SNE is recommended for the inference stage, and the efficiency of the proposed framework is also explained. Given the difficulty of collecting data from real-world environments, the effects of sample imbalance and sample size are also examined. Finally, the approaches employed in the network training to avoid overfitting, and the comparison with related works are detailed.

A. Encoding Strategy and Dimension

Table VI shows the impact of different encoding strategies and encoding dimensions (D) on classification results. Overall,

TABLE VI

ENCODING ABLATION RESULTS ON DATASET SIX-SUBSET UNDER THREE-CLASS CLASSIFICATION OF AF-APB-N

Encoding Strategy	D	Acc.	Pre.	Rec.	F1	Kappa
None	0	0.4808	0.3125	0.4167	0.3556	0.1312
	1	0.5385	0.4780	0.5582	0.4933	0.3055
KPCA-linear	2	0.7500	0.8293	0.7296	0.7358	0.6161
	3	0.8077	0.8765	0.8333	0.8131	0.7143
	1	0.6731	0.4303	0.5714	0.4908	0.4482
KPCA-poly	2	0.7500	0.8302	0.6879	0.7115	0.5878
	3	0.6731	0.7876	0.6877	0.6604	0.5083
	1	0.5000	0.3634	0.4982	0.4057	0.2370
T-SNE	2	0.9615	0.9667	0.9608	0.9623	0.9422
	3	0.6923	0.7128	0.7005	0.6649	0.5355

the encoding strategies are effective, with t-SNE performing better than kPCA. For few-shot learning, having ample prior knowledge, such as sample distributions, is important. The encoding dimension is related to the encoding strategy, and a dimension of two is convenient for planar visualization. In this case, t-SNE best preserves the prior distribution information. KPCA with polynomial and linear kernels shows advantages when the encoding dimensions are 1 and 3, respectively. When the advantages are linked to the encoding branch in the encoding neural network, it implies that low-dimensional representations should have an appropriate nonlinear complexity, while high-dimensional representations should preserve linear complexity as much as possible to avoid interfering with the feature transformations provided by the neural network. The choice of encoding strategy and dimension in practice can be determined after testing based on the complexity of the dataset.

B. Model Deployment and Inference

Inference based on t-SNE requires retraining because it is a global optimization approach and lacks a mapping function. As shown in Fig. 20, parametric t-SNE retains the effectiveness of t-SNE while overcoming the issue of incorporating new samples by a trained mapping function. Based on the principles of parametric t-SNE [26], a simple three-layer multilayer perceptron has been designed for validation, with

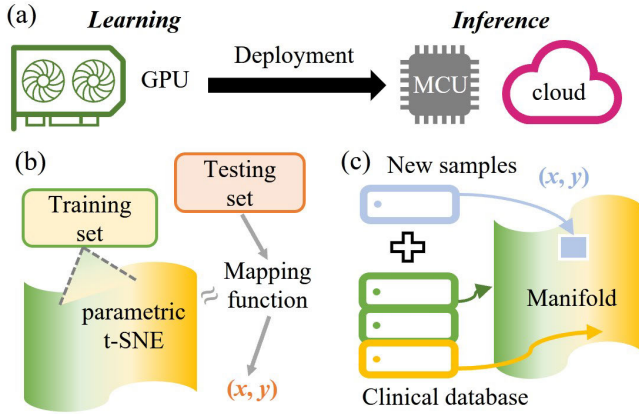


Fig. 20. Inference based on parametric t-SNE. (a) Devices in the learning-to-inference process. (b) Mapping function that parameterizes the t-SNE map. (c) Distribution coordinates of new samples.

TABLE VII
PARAMETRIC T-SNE RESULTS ON SIX-SUBSET DATASET UNDER
THREE-CLASS CLASSIFICATION OF AF-APB-N

Perplexity	Acc.	Pre.	Rec.	F1	Kappa
5	0.9565	0.9468	0.9530	0.9494	0.9328
10	0.9275	0.9269	0.9311	0.9265	0.8911
15	0.9420	0.9293	0.9375	0.9324	0.9099
20	0.9420	0.9433	0.9394	0.9410	0.9124
25	0.9130	0.9167	0.9164	0.9109	0.8697
30	0.8406	0.8885	0.8230	0.8368	0.7522
35	0.7971	0.8339	0.8262	0.7972	0.7016
40	0.8406	0.8492	0.8405	0.8329	0.7604
45	0.8696	0.8895	0.8750	0.8687	0.8055
50	0.8986	0.9049	0.8994	0.8971	0.8480

intermediate dimensions of 256, 128, and 64, respectively, and an output dimension of 2. The corresponding number of parameters (Params) and floating-point operations (FLOPs) of the perceptron are 298 K and 596 K, respectively. The experiments about perplexity in parametric t-SNE have been conducted with strictly separated training and testing sets. In the following training of the classification model, the learning rate is adjusted to $3.5\text{e-}3$, and the early stopping patience threshold is set to 25. Parametric t-SNE avoids potential information leakage during t-SNE experiments, and the consistency of results ensures that findings based on t-SNE retain their reference value.

The results based on parametric t-SNE are shown in Table VII. Results within the 5–25 perplexity range outperform those in the 25–50 range, indicating that the arrhythmia dataset exhibits not a dense distribution but rather several local subcenters, as shown in Fig. 17. The appropriate perplexity value depends on the data density. The greater the data density, the higher the perplexity is recommended. Due to the random initialization of optimization solvers, even when performing the same number of iterations at the same perplexity, the encoding results can vary. One straightforward approach is to run multiple random initializations and select the best one.

TABLE VIII
COMPUTATIONAL COMPLEXITY OF THE FRAMEWORK

Unsupervised stage	Computational complexity	Supervised stage - neural network	Value
ZCT calculation	$O(n)$ [17]	Params	263K
Phase demodulation	$O(n)$ [16]		
T-SNE	$O(n^2)$ [24] $O(n \log n)$ [34]	FLOPs	527K

TABLE IX
DATASET ABLATION RESULTS UNDER THREE-CLASS
CLASSIFICATION OF AF-VPB-N

Dataset	Loss	Acc.	Pre.	Rec.	F1	Kappa
Six-Imbalance $N_s = 185$	CE	0.5405	0.4530	0.4840	0.4102	0.1726
	J	0.6784	0.4419	0.4735	0.4570	0.3616
Six-Subset $N_s = 253$	CE	0.8077	0.8216	0.8083	0.8130	0.7111
	J	0.7308	0.7803	0.7292	0.7345	0.5938
Ruijin $N_s = 297$	CE	0.7466	0.7591	0.7466	0.7405	0.6114
	J	0.6779	0.6763	0.6779	0.6763	0.5051

In the experiments, the accuracies generally exceed 80%, with the best achieving over 90%.

The computational complexity of the proposed framework is shown in Table VIII. The computational complexity of both ZCT calculation and phase demodulation is $O(n)$, indicating that the computation is quite efficient. The Barnes–Hut t-SNE [34] is proposed to accelerate the computation of t-SNE, reducing its complexity from $O(n^2)$ to $O(n \log n)$. During the inference stage, parametric t-SNE is more recommended, and it can be implemented as a multilayer perceptron for efficient computation. The learning framework fully integrates signal-processing methods while considering the extraction of distribution coordinates. This ensures that the computational complexity of the entire process is not overwhelming, even for certain edge-computing MCU.

C. Sample Imbalance and Size

As shown in Table IX, different datasets face varying degrees of sample imbalance and size. In the three constructed datasets, the AF:VPB:N ratios are 1100:666:84, 94:75:84, and 978:757:1239, respectively. The Six-Imbalance dataset suffers from severe sample imbalance. To ensure experimental consistency, the t-SNE encoding strategy is uniformly adopted, with the encoding dimension set to 2.

As shown in Table IX, the difficulty of classification increases with larger sample sizes due to greater complexity in the data distribution, though its impact is not as severe as that caused by sample imbalance. Some loss functions can be employed to mitigate the effects of sample imbalance, but they can also lead to a decrease in accuracy on relatively balanced

TABLE X

ENCODING ABLATION RESULTS ON SIX-SUBSET DATASET UNDER THREE-CLASS CLASSIFICATION OF AF-APB-N

Approach		Acc.	Pre.	Rec.	F1	Kappa
Dropout	✓	0.9615	0.9667	0.9608	0.9623	0.9422
	×	0.6923	0.6918	0.6948	0.6884	0.5320
Epoch	50	0.9615	0.9670	0.9577	0.9614	0.9401
	100	0.9038	0.9015	0.8894	0.8933	0.8529

TABLE XI

ENCODING ABLATION RESULTS ON RUIJIN DATASET UNDER THREE-CLASS CLASSIFICATION OF AF-APB-N

Approach		Acc.	Pre.	Rec.	F1	Kappa
Dropout	✓	0.8919	0.8909	0.8926	0.8915	0.7830
	×	0.8761	0.8808	0.8702	0.8734	0.7474
Epoch	100	0.9099	0.9065	0.9106	0.9082	0.8165
	500	0.5901	0.5649	0.5370	0.5051	0.0808

datasets. Jaccard (J) loss [35] is one of such loss functions, and its definition is as follows:

$$\text{Jaccard Loss} = 1 - \frac{\sum_{i=1}^C s_i \cdot \hat{s}_i + \epsilon}{\sum_{i=1}^C (s_i + \hat{s}_i - s_i \cdot \hat{s}_i) + \epsilon} \quad (28)$$

where C is the total number of classes, s_i is the true label tensor, $\hat{s}_i$ is the predicted label tensor, and ϵ is the smoothing coefficient set to $1e-8$. Compared to CE loss in (28), Jaccard loss maintains a higher focus on small samples, thereby achieving better learning results in an imbalanced dataset.

D. Overfitting Mitigation

When sample sizes are limited, overfitting is likely to occur. In such cases, the dropout approach can mitigate overfitting caused by excessive network complexity. Early stopping approach halts training when the validation set loss ceases to decrease, preventing the network from overfitting to the training set. Empirical evidence from both small-scale and large-scale datasets supports the effectiveness of the approach.

As shown in Table X, on the small balanced dataset with 257 samples, the optimal test accuracy reached 96.15%. Without dropout, overfitting to the training set can cause test accuracy to drop to 69.23%. Similarly, at epoch 100, overfitting leads to a decline in test accuracy.

As shown in Table XI, on the large balanced dataset with 2964 samples, the optimal test accuracy reached 90.99%. Without dropout, overfitting to the training set can cause test accuracy to drop to 87.61%. As the sample size increases, the network complexity becomes better aligned with the data scale, thereby mitigating overfitting caused by excessive network complexity. At epoch 500, overfitting led to a sharp decline in test accuracy, consistent with small sample dataset.

E. Comparison With Related Works

Two state of the art works, mmECG [11] and mmArrhythmia [33], are compared. Based on the waveform reconstruction

TABLE XII

PERFORMANCE COMPARISON WITH RELATED WORKS

Work	Ours	mmECG	mmArrhythmia
Adv.	interpretability; architecture efficiency	domain transformation	end-to-end architecture
Disadv.	additional encoding implementation	interpretability; complexity; data burden	interpretability; complexity
Acc.	0.9615	0.8556	0.9732
Pre.	0.9667	0.5535	0.9460
Rec.	0.9608	0.9762	0.9230
F1	0.9623	0.7075	0.9344

task, the first work offers advantages in domain transformation, thereby providing generalization benefits for unknown tasks. However, it results in a complex network architecture and incurs significant data burden. Additionally, due to the physical inconsistency between electrical and mechanical cardiac activities, interpretability still requires refinement. The second work focuses on solving classification tasks and has designed an end-to-end architecture. However, I/Q signals from different radars, such as CW and FMCW, are not consistent due to differences in hardware design. Directly based on I/Q signals, mmArrhythmia compromises the interpretability of the network input while also failing to address the generalization requirements for inconsistent I/Q signals across different radar architectures. Additionally, the network architecture exhibits considerable complexity to achieve end-to-end prediction.

The proposed neural network requires additional encoding implementation to obtain distribution coordinates, while unified radar displacement serves as the other standard modal input. The architecture of the distribution encoding network facilitates a more physiologically interpretable understanding, and the designed architecture exhibits considerable simplicity in contrast. According to the reported model performance [33], the proposed network has achieved performance comparable to mmArrhythmia and better outperforms on comprehensive metrics such as F1. The advantages (adv.) and disadvantages (disadv.) of these works are summarized in Table XII.

V. CONCLUSION

This article proposes a distribution encoding neural network for contactless cardiac detection of arrhythmia. Four typical arrhythmia classes are discussed, and in a three-class scenario, typical accuracy is around 80%, with a maximum of 96.15% achieved on a balanced small-sample dataset. A workflow employing microwave biomedical radar for clinical data collection and sample observation is detailed. The encoding strategy and dimension encountered by the acquired radar signal samples, along with the impacts of sample imbalance and size, are discussed, which provides beneficial references for related research. Specifically, the proposed processing framework is lightweight and possesses a sound physiological basis for interpretation.

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